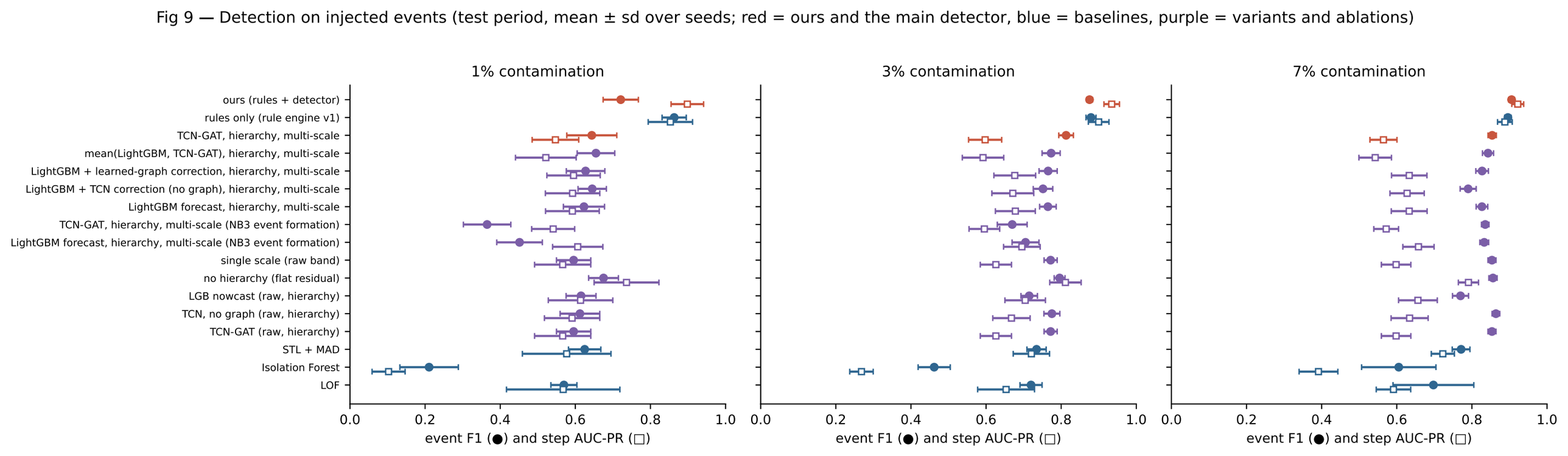
Fig 9 — Detection on injected events (test period, mean ± sd over seeds; red = ours and the main detector, blue = baselines, purple = variants and ablations)
1% contamination
3% contamination
7% contamination
ours (rules + detector)
rules only (rule engine v1)
TCN-GAT, hierarchy, multi-scale
mean(LightGBM, TCN-GAT), hierarchy, multi-scale
LightGBM + learned-graph correction, hierarchy, multi-scale
LightGBM + TCN correction (no graph), hierarchy, multi-scale
LightGBM forecast, hierarchy, multi-scale
TCN-GAT, hierarchy, multi-scale (NB3 event formation)
LightGBM forecast, hierarchy, multi-scale (NB3 event formation)
single scale (raw band)
no hierarchy (flat residual)
LGB nowcast (raw, hierarchy)
TCN, no graph (raw, hierarchy)
TCN-GAT (raw, hierarchy)
STL + MAD
Isolation Forest
LOF
0.0
0.2
0.4
0.6
0.8
1.0
event F1 (●) and step AUC-PR (□)